

19

Rosenbaum-Style p -Values for Matched Observational Studies with Unmeasured Confounding

Rosenbaum (1987b) introduced a sensitivity analysis technique for matched observational studies. Although it works for general matched studies (Rosenbaum, 2002b), the theory is most elegant for one-to-one matching. Different from Chapters 17 and 18, Rosenbaum-type sensitivity analysis works best for matched observational studies for testing the sharp null hypothesis of no individual treatment effect.

19.1 A model for sensitivity analysis with matched data

Consider a matched observational study, with (i, j) indexing unit j in pair i ($i = 1, \dots, n; j = 1, 2$). With exactly matched pairs, unit $(i, 1)$ and unit $(i, 2)$ have the same covariates X_i . Assume IID sampling, and extend Definition 11.1 to define the propensity score as

$$e_{ij} = \text{pr}\{Z_{ij} = 1 \mid X_i, Y_{ij}(1), Y_{ij}(0)\}.$$

Let $\mathbb{S}_i = \{Y_{i1}(1), Y_{i1}(0), Y_{i2}(1), Y_{i2}(0)\}$ denote the set of all potential outcomes within pair i . Conditioning on the event that $Z_{i1} + Z_{i2} = 1$, we have

$$\begin{aligned} \pi_{i1} &= \text{pr}\{Z_{i1} = 1 \mid X_i, \mathbb{S}_i, Z_{i1} + Z_{i2} = 1\} \\ &= \frac{\text{pr}\{Z_{i1} = 1, Z_{i2} = 0 \mid X_i, \mathbb{S}_i\}}{\text{pr}\{Z_{i1} + Z_{i2} = 1 \mid X_i, \mathbb{S}_i\}} \\ &= \frac{\text{pr}\{Z_{i1} = 1, Z_{i2} = 0 \mid X_i, \mathbb{S}_i\}}{\text{pr}\{Z_{i1} = 1, Z_{i2} = 0 \mid X_i, \mathbb{S}_i\} + \text{pr}\{Z_{i1} = 0, Z_{i2} = 1 \mid X_i, \mathbb{S}_i\}} \\ &= \frac{e_{i1}(1 - e_{i2})}{e_{i1}(1 - e_{i2}) + (1 - e_{i1})e_{i2}} \end{aligned}$$

Define $o_{ij} = e_{ij}/(1 - e_{ij})$ as the odds of the treatment for unit (i, j) , and we have

$$\pi_{i1} = \frac{o_{i1}}{o_{i1} + o_{i2}}.$$

Under ignorability, e_{ij} is only a function of X_i , and therefore, $e_{i1} = e_{i2}$ and $\pi_{i1} = 1/2$. Thus the treatment assignment mechanism conditional on covariates and potential outcomes is equivalent to that from an MPE with equal treatment and control probabilities. This is a strategy to analyze matched observational studies we discussed in Chapter 15.1.

In general, e_{ij} is also a function of the unobserved potential outcomes, and it can range from 0 to 1. Rosenbaum (1987b)'s model for sensitivity analysis imposes bounds on the odds ratio o_{i1}/o_{i2} .

Assumption 19.1 (Rosenbaum's sensitivity analysis model) *The odds ratios are bounded by*

$$o_{i1}/o_{i2} \leq \Gamma, \quad o_{i2}/o_{i1} \leq \Gamma, \quad (i = 1, \dots, n)$$

for some pre-specified $\Gamma \geq 1$. Equivalently,

$$\frac{1}{1 + \Gamma} \leq \pi_{i1} \leq \frac{\Gamma}{1 + \Gamma} \quad (i = 1, \dots, n)$$

for some pre-specified $\Gamma \geq 1$.

Under Assumption 19.1, we have a biased MPE with unequal and varying treatment and control probabilities across pairs. When $\Gamma = 1$, we have $\pi_{i1} = 1/2$ and thus a standard MPE. Therefore, $\Gamma > 1$ measures the deviation from the ideal MPE due to the omitted variables in matching.

19.2 Worst-case p -values under Rosenbaum's sensitivity analysis model

Consider testing the sharp null hypothesis

$$H_{0F} : Y_{ij}(1) = Y_{ij}(0) \text{ for } i = 1, \dots, n \text{ and } j = 1, 2$$

based on the within-pair differences $\hat{\tau}_i = (2Z_{i1} - 1)(Y_{i1} - Y_{i2})$ ($i = 1, \dots, n$). Under H_{0F} , $|\hat{\tau}_i|$ is fixed but $S_i = I(\hat{\tau}_i > 0)$ is random if $\hat{\tau}_i \neq 0$. Consider the following class of test statistics:

$$T = \sum_{i=1}^n S_i q_i,$$

where $q_i \geq 0$ is a function of $(|\hat{\tau}_1|, \dots, |\hat{\tau}_n|)$. Special cases include the sign statistic, the pair t statistic (up to some constant shift), and the Wilcoxon signed-rank statistic:

$$T = \sum_{i=1}^n S_i, \quad T = \sum_{i=1}^n S_i |\hat{\tau}_i|, \quad T = \sum_{i=1}^n S_i R_i,$$

where $(R_1, \dots, R_n)$ are the ranks of $(|\hat{\tau}_1|, \dots, |\hat{\tau}_n|)$.

What is the null distribution of the test statistic T under Assumption 19.1 with a general Γ ? It can be quite complicated because Assumption 19.1 does not fully specify the exact values of the π_{i1} 's. Fortunately, we do not need to know the exact distribution of T but rather the worst-case distribution of T that yields the largest p -value under Assumption 19.1. This worst-case distribution corresponds to

$$S_i \stackrel{\text{IID}}{\sim} \text{Bernoulli} \left(\frac{\Gamma}{1 + \Gamma} \right).$$

The corresponding distribution of T has mean

$$E_{\Gamma}(T) = \frac{\Gamma}{1 + \Gamma} \sum_{i=1}^n q_i,$$

and variance

$$\text{var}_{\Gamma}(T) = \frac{\Gamma}{(1 + \Gamma)^2} \sum_{i=1}^n q_i^2,$$

with a Normal approximation

$$\frac{T - \frac{\Gamma}{1 + \Gamma} \sum_{i=1}^n q_i}{\sqrt{\frac{\Gamma}{(1 + \Gamma)^2} \sum_{i=1}^n q_i^2}} \rightarrow N(0, 1)$$

in distribution. In practice, we can report a sequence of p -values as a function of Γ .

19.3 Examples

19.3.1 Revisiting the LaLonde data

We conduct Rosenbaum-style sensitivity analysis in the matched LaLonde data. Using the `Matching` package, we can construct the matched dataset.

```
library("Matching")
library("sensitivitymv")
library("sensitivitymw")
dat <- read.table("cps1re74.csv", header=T)
dat$u74 <- as.numeric(dat$re74==0)
dat$u75 <- as.numeric(dat$re75==0)
y = dat$re78
z = dat$treat
x = as.matrix(dat[, c("age", "educ", "black",
                     "hispan", "married", "nodegree",
```

```

      "re74", "re75", "u74", "u75"]])
matchest = Match(Y = y, Tr = z, X = x)
ytreated = y[matchest$index.treated]
ycontrol = y[matchest$index.control]
datamatched = cbind(ytreated, ycontrol)

```

We consider using the test statistic $T = \sum_{i=1}^n S_i |\hat{\tau}_i|$. Under the ideal MPE with $\Gamma = 1$, we can simulate the distribution of T and obtain the p -value 0.002, as shown in the first subfigure in Figure 19.1. With a slightly larger $\Gamma = 1.1$, the worst-case distribution of T shifts to the right, and the p -value increases to 0.011. If we further increase Γ to 1.3, then the worst-case distribution of T shifts further and the p -value exceeds 0.05. Figure 19.2 shows the histogram of the $\hat{\tau}_i$'s and the p -value as a function of Γ ; $\Gamma = 1.233$ measures the maximum confounding that we can still reject the null hypothesis at level 0.05.

We can also use the `senmw` function in the `sensitivitymw` package to obtain a sequence of p -values against Γ .

```

Gamma = seq(1, 1.4, 0.001)
Pvalue = Gamma
for(i in 1:length(Gamma))
{
  Pvalue[i] = senmw(datamatched, gamma = Gamma[i],
                    method = "t")$pval
}

```

Figure 19.2 shows the plot of the p -value against Γ .

19.3.2 Two examples from Rosenbaum's packages

The `erpcp` dataset is from the R package `sensitivitymw`. It contains $n = 39$ matched pairs of a welder and a control, based on observed covariates age and smoking. The outcome is the DNA elution rate. Figure 19.3(a) shows the histogram of the within-pair differences of the outcomes and the p -value against Γ based on the pair t -statistic. The following R code generates Figure 19.3a.

```

par(mfrow = c(1, 2), mai = c(0.8, 0.8, 0.3, 0.3))
data(erpcp)
hist(erpcp[, 1] - erpcp[, 2], main = "erpcp",
     xlab = expression(hat(tau)[i]),
     freq = FALSE)

Gamma = seq(1, 5, 0.005)
Pvalue = Gamma
for(i in 1:length(Gamma))
{
  Pvalue[i] = senmw(erpcp, gamma = Gamma[i], method = "t")$pval
}
gammastar = Gamma[which(Pvalue >= 0.05)[1]]

```

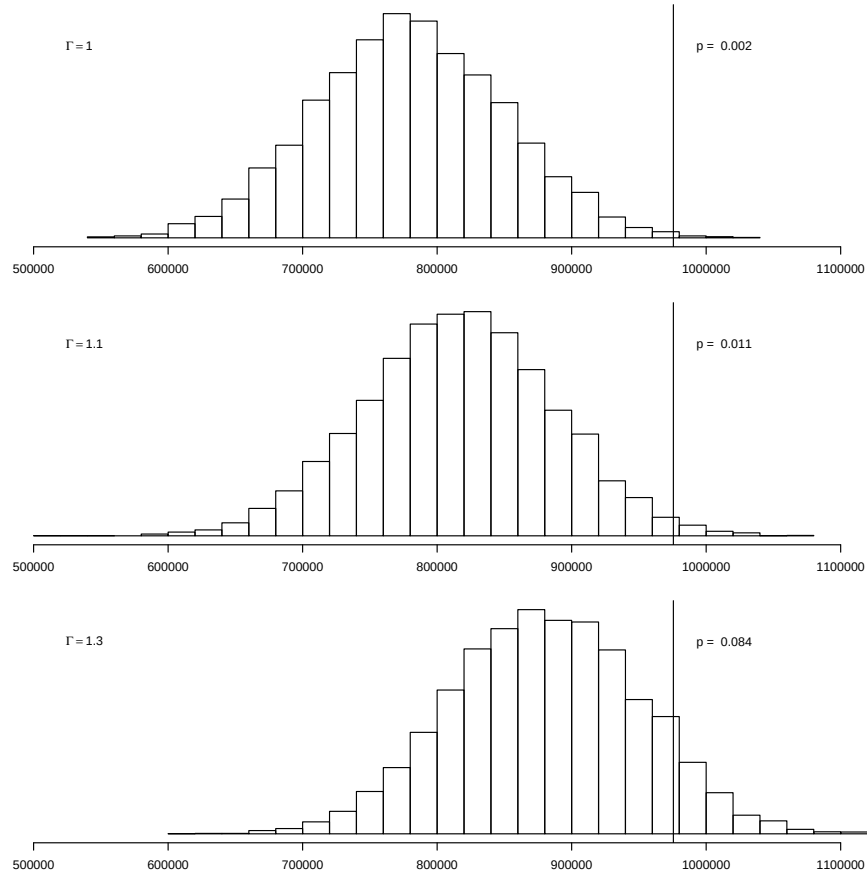


FIGURE 19.1: The worst-case distributions of $T = \sum_{i=1}^n S_i |\hat{\tau}_i|$ with S_i IID Bernoulli($\Gamma/(1 + \Gamma)$), based on the matched LaLonde data.

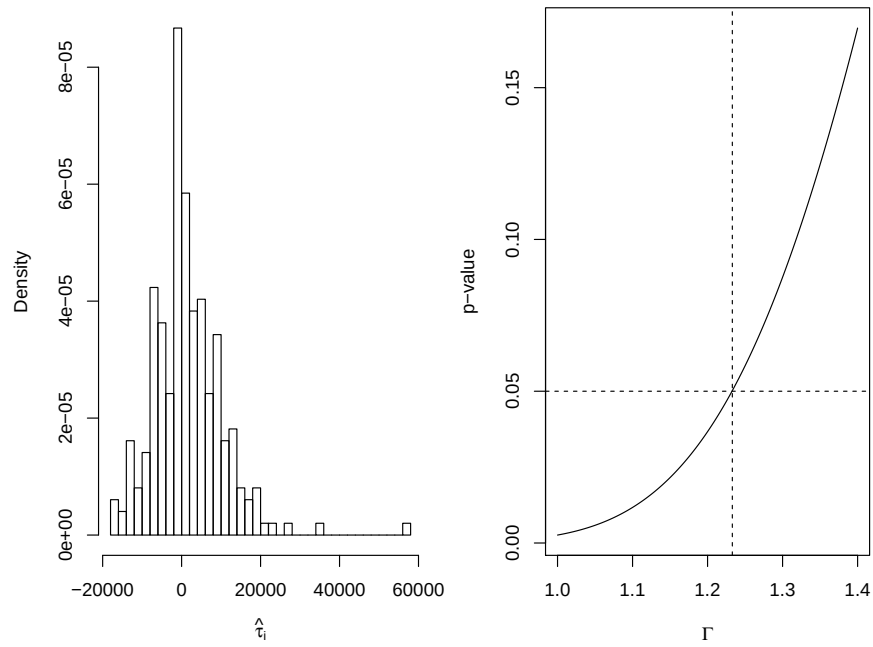


FIGURE 19.2: p -value as a function of Γ , based on the matched LaLonde data.

```

gammastar

plot(Pvalue ~ Gamma, type = "l",
     xlab = expression(Gamma),
     ylab = "p-value")
abline(h = 0.05, lty = 2)
abline(v = gammastar, lty = 2)

```

The `lead250` dataset is from the R package `sensitivitymw`. It contains $n = 250$ matched pairs of a daily smoker and a control of nonsmoker, based on observed covariates gender, age, race education level, and household income from NHANES. The outcome is the blood lead level in $\mu g/l$. Figure 19.3(b) shows the histogram of the within-pair differences of the outcomes and the p -value against Γ based on the pair t -statistic. The following R code generates Figure 19.3b.

```

par(mfrow = c(1, 2), mai = c(0.8, 0.8, 0.3, 0.3))
data(lead250)
hist(lead250[, 1] - lead250[, 2],
     main = "lead250",
     xlab = expression(hat(tau)[i]),
     freq = FALSE)

Gamma = seq(1, 2.5, 0.001)
Pvalue = Gamma
for(i in 1:length(Gamma))
{
  Pvalue[i] = senmw(lead250, gamma = Gamma[i], method = "t")$pval
}
gammastar = Gamma[which(Pvalue >= 0.05)[1]]
gammastar

plot(Pvalue ~ Gamma, type = "l",
     xlab = expression(Gamma), ylab = "p-value")
abline(h = 0.05, lty = 2)
abline(v = gammastar, lty = 2)

```

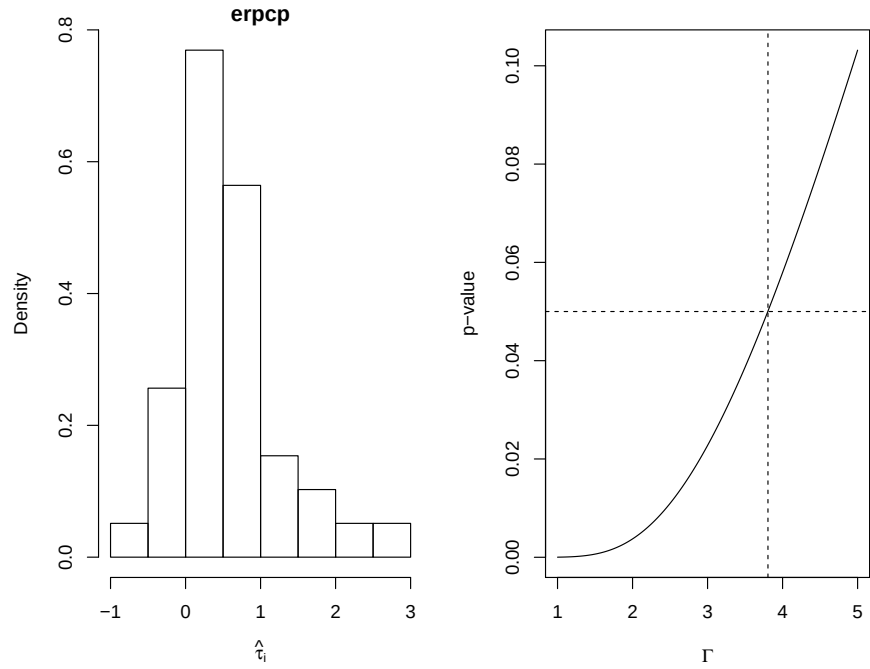
19.4 Homework Problems

19.1 A model for Assumption 19.1

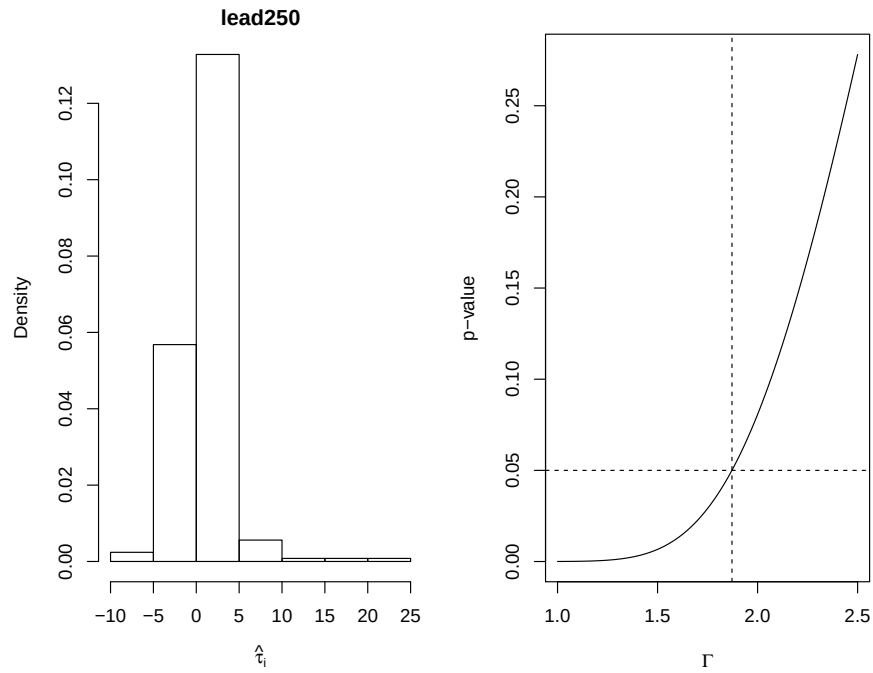
Assumption 19.1 has the following equivalent form.

Assumption 19.2 *The propensity score satisfies the following model*

$$\log \frac{\pi_{ij}}{1 - \pi_{ij}} = g(X_i) + \gamma U_{ij}, \quad (i = 1, \dots, n; j = 1, 2)$$



(a) erpcp data



(b) ead250 data

FIGURE 19.3: Two examples: histogram of the $\hat{\tau}_i$'s and p -value as a function of Γ

where $g(\cdot)$ is an unknown function and $U_{ij} \in [0, 1]$ is a bounded unobserved covariate for unit (i, j) .

Show that if Assumption 19.2 holds, then Assumption 19.1 must hold with $\Gamma = e^\gamma$.

Remark: Rosenbaum (2002b, page 108) also shows that if Assumption 19.1 holds, then Assumption 19.2 must hold for some U_{ij} 's. Assumption 19.2 may be easier to interpret because γ measures the log of the conditional odds ratio of U on the treatment; see Chapter B.6 for the interpretation of the coefficients in the logistic regression.

19.2 Application of Rosenbaum's approach

Re-analyze Example 10.3 using Rosenbaum's approach based on matching.

19.3 Recommended reading

Rosenbaum (2015) provides a tutorial for his two **R** packages for sensitivity analysis with matched observational studies.